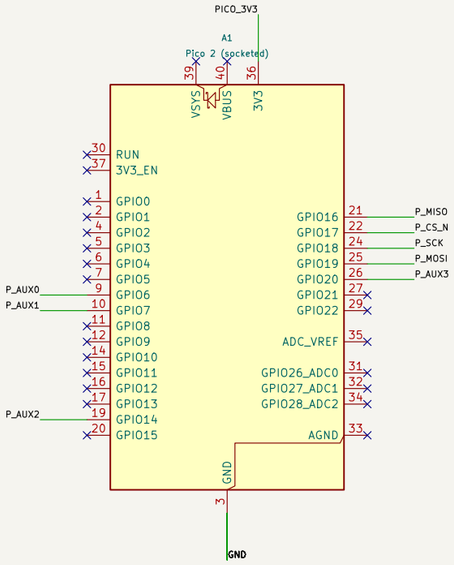
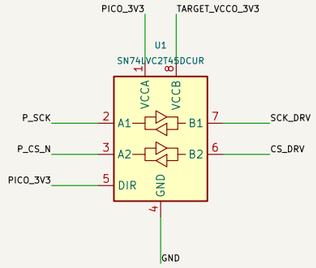
RP2350 FPGA INTERFACE / DEBUG — REV 1 CANDIDATE

DRAFT / NOT FOR FABRICATION — independent 3.3 V domains; dedicated SPI mode 0

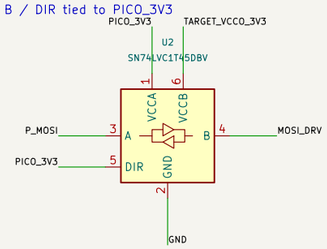




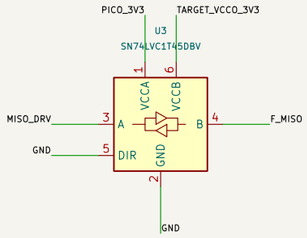
U1: A -> B / DIR tied to PICO_3V3



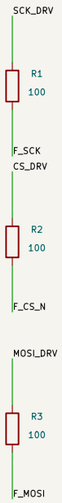
U2: A -> B / DIR tied to PICO_3V3

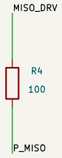


U3: B -> A / DIR tied to GND

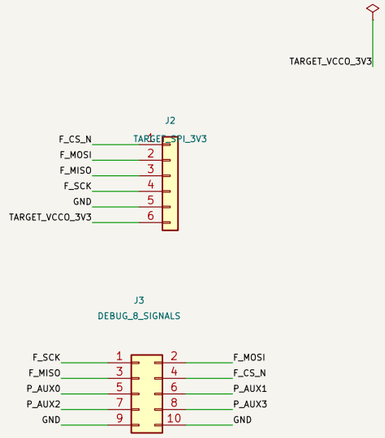








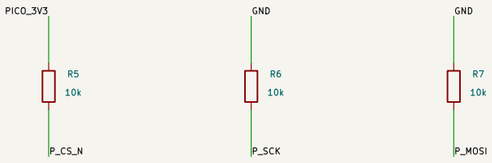


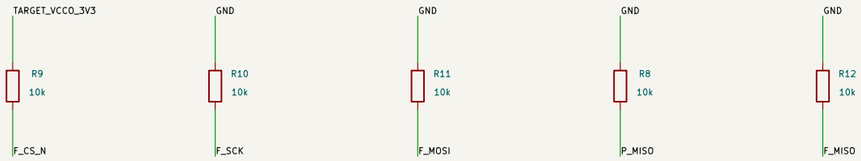


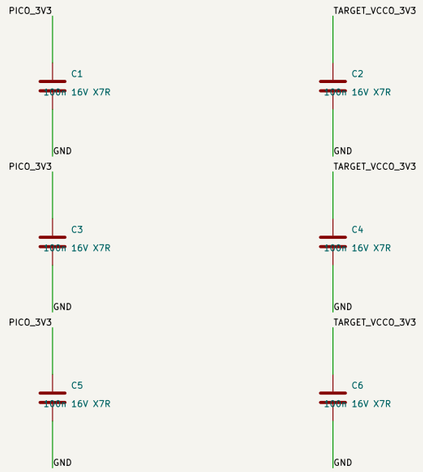
J3: LOGIC ANALYSER / DEBUG — 8 SIGNALS

No rail tie. Common ground must connect first.

TARGET rail powers VCCB; not sense-only.







R1–R4: place at translator outputs. 100 ohm is a starting value.

Bypass pairs: C1/C2 U1; C3/C4 U2; C5/C6 U3.

Place at IC supply pins; short ground returns.

Pico USB powered; VBUS / VSYS / 3V3_EN / RUN unused on carrier.

A1 footprint: two 1x20 sockets; verify fit before layout.

Idle bias — each net uses its own domain only

Power-off isolation candidate; no UVLO / power-good / OE.

Brownout transaction blocking is NOT implemented.

No UART / flash / other cross-board bypass paths included.

No bench qualification; VCCO identity and ramp tests outstanding.

J2 target connector: mapped cable required.

1 CS# | 2 MOSI | 3 MISO | 4 SCK

5 GND | 6 TARGET_VCCO_3V3 (input)

Verify target voltage, bank rail and pinout.

J3 SPI: target domain; AUX0–3: Pico domain.

AUX0–3 unprotected; never bypass U1–U3.

Rev 1 power: USB -> Pico regulator; no extra regulator.

Pico external 3V3 budget: <=100mA initial target.

Planning slots (mA), NOT measured:

Interface 10 | INA226 5 | I2C sensors 10

Storage 50 | LEDs 2 | Margin 23 = 100

Modules TBD; check peak/startup current before fitting.

Target VCCO: B-sides only; never peripheral power.

NOT FOR FABRICATION; no brownout threshold

Sheet: /
File: southbridge_rev1_candidate.kicad_sch

Title: RP2350 FPGA interface / debug — Rev 1 candidate

Size: A3	Date: 2026-11-05	**Rev: R1-CANDIDATE**
KiCad E.D.A. 10.0.7		Id: 1/1